

Systems Thinking

Finding Leverage Points: Reversing Vicious Cycles

Help communities understand what's driving ecological decline and where a single well-chosen action can reverse it.

About This Tool

Most environmental problems in rural communities aren't caused by one thing. They're the result of interconnected pressures that reinforce each other: declining fish stocks push fishers toward destructive methods, which damages reefs further, which reduces fish stocks more. These are vicious cycles, and they're why problems that seem straightforward resist simple fixes. Systems thinking gives communities a way to see those cycles clearly. Once the pattern is visible, the question shifts from "how do we fix everything?" to "where can we intervene to start reversing it?" That's a question communities can actually answer.

How This Tool Works

The four parts of this tool are structured workshops. You'll run them in sequence with your community, building understanding at each step before moving to the next. Each session produces a specific output: a list of losses and causes, a diagram of the vicious cycle, a set of evaluated interventions, a plan for sustaining what works. The community sessions that follow draw directly on this work. The systems thinking process is how you and the community build the shared picture that makes those sessions possible.

The Four-Part Process

The four parts build on each other. Don't skip ahead.

Recognize It

Gather the community's story of decline through small-group conversations and a community forum. You'll come out with a list of losses and causes, and a rough timeline of how things unfolded.

Map It

Bring the community together to turn that list into a visual diagram. You'll trace the feedback loops connecting losses and causes until the vicious cycle is visible on paper.

Reverse It

Look at the diagram and identify where you can realistically intervene. Evaluate options together and choose a starting point that's achievable and likely to build momentum.

Lock It In

As initial interventions take hold, map the new situation. Identify what's working, strengthen it, and build the economic and social connections that make recovery self-sustaining.

Who Holds the Knowledge

The knowledge is already in the community. Elders remember what the land and water looked like before the decline began. Women often hold detailed knowledge of how the ecosystem functions day to day. Young people understand current conditions and pressures. The systems thinking process draws that knowledge out and organizes it into a shared picture no single person could build alone.

Recognize It

Gather the community's story of decline and build a shared list of losses and causes that you'll map in the next step.

Your Goal

Before you bring the full community together to map the problem, you need to understand the situation well enough to facilitate that conversation. Recognize It is fieldwork. You're building a picture of what happened, when, and to whom. The mapping session starts from shared story, not a blank page.

You'll come out of this step with two things: a rough timeline of how the decline unfolded, and a working list of losses and causes that the community will organize and diagram in Map It.

Start with Smaller Groups

Don't begin with a community-wide forum. Meet first with smaller common-interest groups: fishing families, farmers, women's groups, elders. These conversations are easier to have honestly. People speak more directly in smaller settings, and you'll hear things that won't come up in a larger group.

Aim for broad representation across these early conversations. You're not looking for consensus yet. You're listening for the range of experience.

Walk the Environment

If possible, walk the affected land or water with community members. The environment itself prompts memory. People recall specific losses when they're standing where they happened. Ask what used to be here. Ask what changed. Let them tell the story in place.

Listen First

Your role in this phase is to draw out the community's own account of what happened. Avoid offering explanations or solutions. Ask questions, then listen. The story needs to come from them.

Useful questions:

- What did this place look like before things started to decline?
- When did you first notice something was wrong?
- What do you think started it?
- What happened after that?
- How has it affected your household? Your income? Your work?

For each problem that surfaces, follow it: what did this lead to? What else got worse because of it?

What You're Listening For

As you gather stories, keep these patterns in mind as a guide for when you have enough material to move on.

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The Situation: Community members could see interconnected problems: declining fish catches, damaged coral reefs from dynamite and cyanide fishing, and increasing poverty.

A trigger or starting point.

Was there a specific event or change that set things in motion? A new technology introduced, a policy change, a shift in land use or ownership?

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The Trigger: The introduction of destructive fishing methods created a vicious cycle:

Destructive fishing methods → Reef damage → Fewer fish → More destructive methods →
↻ More reef damage

A sequence of connected problems.

Does one problem seem to have caused another? You're looking for chains: this led to that, which made this worse, which caused that.

On Apo Island

Amplifying Cycles: can reinforce existing vicious cycles and create new ones:

Fishing further away → Less incentive to protect local waters → More destructive methods
→ More reef damage → ↻ Even fewer fish

A shift from a better past.

Can elders describe conditions before the decline? That contrast sharpens the story and helps identify what the community is trying to recover.

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The Memory: Elders remembered when the reefs were healthy and fish were abundant.

Thresholds or turning points.

Moments when things got noticeably worse, or faster. These often mark where the system crossed into a new state.

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The Threshold: Fishermen could no longer make a living from island fishing grounds and had to travel farther from home.

Impacts on livelihoods.

For each environmental loss, what has it meant for people's lives, income, and work? These connections matter for building shared motivation in the mapping session.

On Apo Island

The Crisis: Eventually, there was nowhere close by with adequate fish, and the community faced collapse.

What You're Building

As you work through these conversations, keep a running list. Two columns is enough: losses and causes. Don't worry about organizing it yet. That's Map It's job. You're collecting material.

By the time you convene the community forum, you should have a rough timeline of the decline and enough documented losses and causes that the mapping session has real content to work with from the start.

The Community Forum

Once you've completed your smaller-group fieldwork, bring the broader community together. The goal is to arrive at shared consensus on the story of decline. You're not generating the list from scratch here; you're validating, expanding, and agreeing on what you've gathered.

The forum ends when there's broad agreement on the losses, the causes, and roughly how they connect. The next step is to diagram that.

Map It

Create a visual diagram of the feedback loops behind your community's decline.

Understanding Your Vicious Cycles

This section provides step-by-step guidance for bringing the community together to understand what's causing their problems. You'll create a boxes-and-arrows diagram showing the feedback loops that have been driving decline.

Convene Your Mapping Group

Who to Include

- Elders who remember how the situation developed over time
- Women who often hold passed-down knowledge and may observe different aspects of community life
- Young people who understand current realities and possibilities
- People from different roles in the community (farmers, fishers, merchants, etc.)
- If you're an outside NGO, include diverse community members in this process
- People with a history of community service

How to Convene

- Meet in a neutral, accessible space where all participants feel comfortable
- Plan for 2-3 hours for initial mapping session
- Ensure language and format work for all participants
- Consider whether mixed-gender or separate groups work better in your context

Ingredient: Building Shared Vision

Gathering diverse community members to understand your situation together builds shared vision and commitment, not only analysis. When people participate in mapping the problem, they develop a common understanding and ownership of solutions.

Start With the Story

First Question

Think back to the very first action or activity that set the decline in motion. This is what we call the "negative tipping point lever": the trigger that started things going in the wrong direction.

Additional Guiding Questions

- What was different before this change happened?
- When did you first notice things getting worse?
- What else changed around that time?

Process

- Have participants share stories and memories
- One person should capture key points (use large paper visible to all, or Post-it notes)
- Look for: triggering events, changes in behavior, environmental shifts, outside pressures

- Don't worry about getting everything perfect; you'll refine as you go

On Apo Island

Elders remembered when fish were plentiful and coral reefs were healthy. The trigger was the combination of population growth and the introduction of destructive fishing methods like dynamite and cyanide fishing in the 1970s and 1980s.

Map the First Feedback Loop

Key Question

What happened in the ecosystem or community to form a circular chain of effects (a feedback loop) that kept the decline going?

Process

- Start with the trigger/negative tipping point lever you identified
- Ask: What did this cause or make worse?
- Then: What did that cause or make worse?
- Continue until you complete a circle back to something that reinforces the original problem
- Draw this as boxes (the problems/factors) connected by arrows (showing what affects what)

The boxes in your diagram represent:

- The tipping point lever (the trigger)
- The "bottom line" (the main problem you're trying to solve, like declining fish stocks or degraded forest)
- Any ecosystem or community elements that complete the feedback loop

Arrows between boxes show what affects what.

On Apo Island — first loop

Destructive fishing methods → Coral reefs damaged → Fish breeding grounds destroyed → Declining fish stocks → Fishers use even more destructive methods → ↻ Destructive fishing methods

Add Reinforcing Feedback Loops

Key Question

What happened after that to intensify the decline even more?

Process: Look for additional circular chains of effects that reinforced the first feedback loop. Add more boxes and arrows to your diagram to show how the decline expanded or accelerated.

You may find:

- Second-order effects (the decline caused other problems that made things worse)
- Social dynamics that reinforced environmental decline
- Economic pressures that developed as a result
- Multiple reinforcing loops working together

Keep mapping: Continue adding boxes and arrows until you've captured the major dynamics driving your decline.

On Apo Island — reinforcing loops

The first loop was reinforced by economic and social dynamics:

Declining catches → Lower income → More economic pressure → More intensive/destructive fishing → Further reef damage → ↻ Declining catches

And another loop:

Fishers traveling farther → Less time protecting local waters → Reduced community cohesion → Weaker informal rules → More destructive fishing → ↻ Fishers traveling farther

What You've Created

You now have a shared visual representation of the system causing your problems — your main tool for spotting where interventions will be most effective.

Next Steps

You now understand the vicious cycles driving decline. Next, identify where you can intervene to reverse them.

Reverse It

Identify leverage points and design interventions that can reverse vicious cycles into virtuous ones.

Breaking the Cycle

You understand the vicious cycles driving decline. This step is about identifying where you can intervene to reverse them — finding actions that can break the negative feedback loops and start positive ones.

Step 1: Identify Potential Leverage Points

Look at your vicious cycle diagram and ask:

Where could you intervene?

- Which connections between problems could be broken?
- Which elements could be replaced with positive alternatives?
- Where is the community ready and able to act?
- Are there examples from other communities that succeeded?

What resources or support do you have access to?

- Community skills and knowledge
- Natural resources that could regenerate
- Traditional practices that worked in the past
- Outside support (NGOs, government programs, etc.)
- Economic opportunities tied to restoration

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Step 1: Leverage Point

The community identified that protecting a portion of their reef could allow fish populations to recover, which would then spill over into fishing areas.

Ingredient: Community Resolve

The evaluation process reveals whether you have the resolve needed. If community support seems low for a crucial intervention, that's valuable information; you may need to build resolve first through education and dialogue.

Step 2: Envision the Positive Flip

For each potential intervention point, imagine what success would look like:

- If this intervention works, what positive changes would ripple through the system?
- Could it start new positive feedback loops?
- How would the boxes and arrows in your diagram change?

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Step 2: The Vision

They envisioned creating a marine sanctuary on 10% of their fishing grounds.

Fish stocks recover → More fish spill over to fishing areas → Catches improve →
Community sees benefits → More support for protection → Stronger enforcement → ↺
Better recovery

[virtuous cycle]

Step 3: Evaluate Feasibility and Impact

The positive tipping point lever consists of the actions taken to reverse those elements plus the environmental technology used for those actions and the community organization to make the technologies and actions effective.

For each potential intervention, work through these questions as a group:

Impact Questions

- How big a change could this make?
- Does it address a root cause or just a symptom?
- Will social and environmental gains go hand in hand?
- Could it trigger positive reinforcement?
- Is it strong enough to reverse the bottom line?

Tool Download: Impact Worksheet

Quantify the impact of your interventions

Feasibility Questions

- What resources would this require? (money, time, skills, materials)
- Do we have community support for this?
- What obstacles might we face? (social, political, economic)
- Do we have (or can we get) what we need to try this?

Risk Questions

- What could go wrong?
- Are there unintended consequences we should consider?
- Is this reversible if it doesn't work?
- Will there be pushback from other entities?
- Will there be encroachment or takeover from other entities?

Tool Download: Feasibility Worksheet

Understand the challenges, risks, and their impact

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Step 3: Evaluation

Impact: Could reverse the fundamental problem (declining fish stocks)

Feasibility: Drew on traditional decision-making practices; representatives from all families discussed until reaching consensus

Resources: Created volunteer Marine Guard; established clear rules

Step 4: Choose Your Starting Point

You don't need perfect information to begin. Based on your evaluation:

- Which 1-3 interventions seem most promising?
- Is there a logical sequence? (some things need to happen before others)
- Can you start with something achievable to build momentum?

Remember

- You can try multiple interventions, especially if they reinforce each other
- Starting with a smaller, achievable win can build the community resolve needed for bigger changes
- You'll learn by trying. Your understanding will improve as you act

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Step 4: Starting Point

They began with the marine sanctuary and community management, building on their existing consensus-based decision-making traditions.

Next Steps

Once you've started to see success, the final step is to lock in those gains by building mutually reinforcing positive cycles.

Lock It In

Sustain and Scale: Design for lasting change that reinforces itself and spreads to other communities.

Building Mutually Reinforcing Success

As ecosystem restoration proceeds, the next step is to lock in the gains and increase the sustainability of restoration through new enterprises that reinforce ecosystem restoration and integrate it with the social fabric of the community.

Key Concept

Early success can die out if it's not built into the fabric of community life. Sustainability comes from multiple mutually reinforcing cycles.

Guiding Questions

- What new opportunities does initial success create?
- How can benefits be shared broadly to build ongoing support?
- What new activities or enterprises could build on this foundation?
- How can this become "just the way things are done" rather than a special project?

Process

- Revisit this 6-12 months after initial interventions
- Map the new positive cycles that are emerging
- Identify opportunities to strengthen and multiply them
- Look for ways to institutionalize success (community rules, new traditions, ongoing activities)

Ingredient: Success Breeds Success

This is where the "success breeds success" dynamic becomes crucial. Initial wins create opportunities, resources, and motivation for further positive change.

Making Change Permanent

The goal is to shift from a special project requiring constant attention to a new normal that sustains itself through multiple reinforcing mechanisms. This might include:

- Economic enterprises that depend on and support restoration
- Educational programs that build pride and understanding
- New community traditions or celebrations
- Recognition and support from outside organizations
- Clear rules and enforcement mechanisms that have community buy-in

On Apo Island

Initial success (fish recovery) led to multiple mutually reinforcing cycles:

Economic Cycle

Marine tourism income → Funds for community → More resources for protection → Better enforcement → Healthier reefs → ☞ More tourism

Educational Cycle

Educational programs → Pride in achievement → Stronger community identity → More support for protection → Better outcomes → ☞ More to teach about

Recognition Cycle

Scientific recognition → Outside support → Enhanced credibility → More resources → Better monitoring → ☞ More recognition

The Result: These multiple reinforcing cycles made the marine sanctuary permanent, not just a temporary project. It became integrated into the community's identity and economy.

Congratulations!

You've completed the Systems Thinking tool. Remember that this is an iterative process. As you take action and learn, you'll refine your understanding and adjust your approach. The cycle of recognizing, mapping, reversing, and locking in continues as conditions change.